

**UConn**

School of Business

# Lab 1 — First Commit

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**Tonight in 20 steps · simple edition**

OPIM 5512 · Module 1 · Data Science using Python

**The deliverable tonight is a REPO, not a notebook.**

# The deal

You and a partner each own a DIFFERENT half of the same story: one takes the weather, one takes the electricity demand.

The data is already clean. You run a line plot, write ONE histogram, and get both into a shared repo — through a branch, a pull request, and your partner's review.

Then, together, you fill in a one-page report where the plots are already wired in.

No cleaning, no modeling tonight — those are coming. Week 1 is the workflow you'll use every week.

 **No AI tonight. Your only code is a three-line histogram — type it.**

You're about to approve a partner's code. Approving code you didn't read is the habit that makes review worthless.

# Stack check

**GitHub account** + GitHub Desktop installed and signed in

**Google Colab** opens in your browser

**Your campus word** — hartford or stamford (one variable in the weather notebook)

**Have the instructions open** beside Colab — every click is in there, in order

**Pick roles now** — Partner A = weather, Partner B = demand

Odd one out pairs with Dave. Online / solo? Two accounts, or pair over Teams.

# Make the shared repo — from the template

- 1 · Pair up — Partner A (weather) / Partner B (demand)
- 2 · Partner A: template repo → Use this template → new PUBLIC repo, owner = you
- 3 · Partner A: Settings → Collaborators → add B → B accepts the invite
- 4 · Partner A: Settings → Rules → ruleset on main: require a PR + 1 approval
- 5 · Both: clone the repo ONCE in GitHub Desktop
- 6 · Each: New branch (dev-weather / dev-demand) → Publish branch

**The template already holds the clean data, both notebooks, images/, README and REPORT. You build nothing by hand.**

# Run it, write one histogram, ship it

- 7 · Colab → File → Open notebook → GitHub tab → paste YOUR repo URL → open your notebook
- 8 · A sets CAMPUS. Runtime → Run all. The line plot appears (its PNG is already saved)
- 9 · In the TODO cell: write your histogram (the shape is given). Run it. Title + units.
- 10 · Run the download cell → two PNGs land in Downloads
- 11 · File → Save the notebook → your repo · your dev- branch · same path
- 12 · Desktop → Show in Explorer → drag the PNGs into images/ → Commit → Push

**Rename any '(1)' file before you drag — the report links to the exact names. Plain Save, not Ctrl+S.**

# The two-way loop

13 · Compare & pull request → Create → Reviewers → your partner

14 · Open your PARTNER's PR → Files changed → read their histogram, look at the PNGs → Approve

15 · Merge both PRs → Delete branch

16 · Both: Desktop → main → Fetch → Pull — four PNGs and both notebooks are now on your laptop

**You can't approve your own PR — that's the gate. Approve only what you actually read.**

# Say what the plots say

- 17 · One screen: REPORT.md → Edit → one sentence per → line (every number gets a unit)
- 18 · Commit to a new branch 'report' → PR → the other partner approves → Merge
- 19 · Ahead of schedule? Run the joint notebook → temp\_vs\_load.png → images/ → report section 5
- 20 · Read-out: one plot on screen, one sentence — check Insights → Network shows the loop

**Flying? Both add a line to README → Findings on separate branches, merge one, then the other: your first merge conflict.**

# Don't panic

**Commit early.** Once something's committed, it's essentially impossible to lose.

**Fetch & Pull can't hurt you.** Fetch = peek, Pull = download. Neither overwrites your work.

**Same path every save** — a different path makes a second notebook, not an update.

**main rejecting your push is the protection working** — you're on main when you should be on your branch.

**PNG didn't download?** The download cell names the missing file — run the cell that makes it, then re-run.



# Online / solo students

**Pair over Teams if you can** — each on your own account, one owns the repo and adds the other.

**Otherwise, two accounts** (needs a second email) — play both A and B, including approving as the other account.

**Simplest solo** — one account with Required approvals = 0: branch → PR → merge yourself.

**Either way** — join the live session on Teams, and Friday office hours are highly recommended.

# Definition of done

- ☐ Both partners are collaborators; branch protection on main
- ☐ images/ has FOUR PNGs with the exact filenames (two per partner)
- ☐ Both notebooks saved back with a histogram in each
- ☐ REPORT.md — one real sentence under each plot, plus one honest caveat
- ☐  $\geq 3$  merged pull requests (one each + the report), branches deleted, both authoring AND reviewing
- ☐ A network graph showing the loop going both ways